

Talha Quddoos

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PROFESSIONAL SUMMARY

PEC-registered engineer and M.E. candidate in Transportation Engineering (Utah State University, Dec 2026) with 4+ years of research and consulting experience spanning transportation planning, traffic safety, GIS, and infrastructure resilience. EIT-eligible, sitting for the FE Civil exam in September 2026. Brings a rare combination of field engineering, geospatial analysis, and peer-reviewed research to public agency, consulting, and research roles. Authorized to work in the U.S. under OPT — no employer sponsorship required.

SKILLS

Transportation & Civil Engineering: Transportation Planning, Travel Demand Modelling, Traffic Safety Analysis, Transit Planning & Operations (GTFS), Crash Hotspot Analysis (KDE), Drone Surveying, Geospatial Engineering, Stakeholder & Public Agency Coordination

Software & Tools: PTV Vissim, OpenPaths Cube, SUMO, ArcGIS Pro, ArcGIS Desktop, QGIS, AutoCAD, PostgreSQL/PostGIS, Google Earth Engine, Mapbox GL JS, Leaflet, Agisoft Metashape, Global Mapper, PCI Geomatica

Programming & Analysis: Python, R, SQL, JavaScript, Statistical Modeling, Jamovi

EDUCATION

Utah State University

*M.E., Civil and Environmental Engineering (Transportation Engineering)
Recipient, USU Climate Adaptation Science (CAS) Scholarship: \$5,000.*

Logan, UT

Graduating Dec 2026

University of the Punjab

M.Phil., Space Sciences (Spatial Science and Remote Sensing)

Lahore, Pakistan

2022

University of Engineering and Technology (UET) Lahore

B.Sc., Transportation Engineering.

Lahore, Pakistan

2020

CERTIFICATIONS

- Fundamentals of Engineering (FE) Civil Exam — Scheduled September 2026
- Registered Engineer, Pakistan Engineering Council (PEC) — Transport/399

EXPERIENCE

Teaching Assistant

Utah State University, Dept. of Civil and Environmental Engineering

Sept 2026 – Dec 2026

Logan, UT

- Assisted in instructing a graduate/undergraduate Transportation Planning course covering travel demand modeling, including four-step modeling and network coding in Cube.
- Supported students in building and calibrating a regional travel demand model, including trip generation, distribution, mode choice, and assignment steps.
- Held office hours, graded assignments, and guided students regarding Cube software workflows.

Graduate Research Assistant

Utah State University, Dept. of Civil and Environmental Engineering

Jan 2025 – Aug 2026

Logan, UT

- Gathered emergency managers' perspectives on the safety of wireless electrified roadways in disaster-prone states across the U.S., supporting infrastructure resilience planning. (ASPIRE Research Center)
- Applied GIS and statistical analysis to assess electric vehicle infrastructure resilience under wildfire risk in the US. (ASPIRE Research Center)
- Performed coverage and equity analysis on San Francisco Muni Bus Service using schedule data (GTFS).
- Led a temporal study on mobility disruption and wellbeing following the 2023 Vermont floods - presented at AAG Annual Meeting 2026.

- Assessed infrastructure systems risk to bridges in rural wildland–urban interface areas exposed to wildfire, integrating structural and sociodemographic data - presented at the ASCE Architectural Engineering Institute & Infrastructure Resilience Division (AEI IRD) Conference 2026.
- Applied geospatial artificial intelligence methods to detect post-earthquake structural damage from satellite imagery, supporting rapid damage-assessment workflows – presented at AI Summit UoU 2025.
- Contributed to the Cache County Climate Assessment through the USU Climate Adaptation Science Cohort, informing local climate adaptation planning. (Climate Adaptation Science, USU)

Collaborations

Aspire Research Center, PHI Lab - University of Oxford, University of Vermont, Climate Adaptation Science USU, The Nature Conservancy.

Research Assistant

Feb 2023 – Dec 2024

Center for Urban Informatics, Technology and Policy, LUMS University

Lahore, Pakistan

- Built an AI-driven WebGIS route optimization dashboard for a municipal waste management company, reducing fuel consumption by 100,000 to 150,000 liters through optimized vehicle routing.
- Identified traffic crash hotspots using kernel density estimation (KDE+) and optimized ambulance staging locations to improve emergency response coverage.
- Assessed population and infrastructure exposure for a proposed elevated expressway using AI-extracted building footprints and WorldPop density data in QGIS.
- Developed an AI-based smart traffic management system integrating real-time computer vision with traffic modeling software for a historic urban district.

Collaborations

Walled City Authority Lahore, Punjab Safe Cities Authority, Emergency Services Department (Rescue 1122), Lahore Waste Management Company, PHI Lab - University of Oxford, Beaconhouse National University (BNU)

GIS & Remote Sensing Expert

Oct 2022 – Jan 2023

SheherSaaz (Urban Planning Consultants)

Lahore, Pakistan

- Provided GIS and remote sensing analysis to support urban planning and land-use projects.

Transportation Engineer

Jan 2022 – Feb 2022

Punjab Mass Transit Authority – Speedo Bus Service

Lahore, Pakistan

- Supported planning and operations of a public transit feeder bus network, coordinating with transit authority stakeholders.

Additional Experience

- Research Student, The Spatio – Geospatial Engineering Consultants (Dec 2021 – Aug 2022)
- Research Student, Dept. of Transportation Engineering, UET Lahore (Sept 2019 – Aug 2020)

PUBLICATIONS & PRESENTATIONS

- 4 peer-reviewed publications and 5 conference presentations on transportation safety, climate resilience, and geospatial analysis. Venues include AAG, ASCE AEI, NeurIPS Workshop, ICONIP, and IMECE. Full list available at talha-q.github.io.

PROFESSIONAL AFFILIATIONS & ACTIVITIES

- Member, Institute of Transportation Engineers (ITE) – USU Student Chapter
- Member, American Association of Geographers (AAG)
- Delivered multiple technical training workshops on GIS and data-driven road safety tools.
- Represented ITE USU at the Traffic Bowl, ITE Mountain District Conference, 2025 (Santa Fe) and 2026 (Flagstaff)